GRIDPRINT analysis report

Verdict

VERIFIED

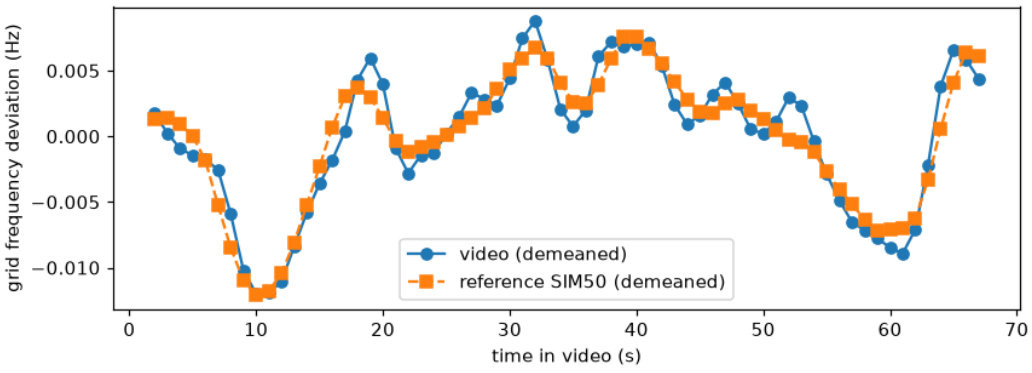
A grid signal is present and matches the reference log at one time. No significant edit evidence was found.

Report state: FINAL

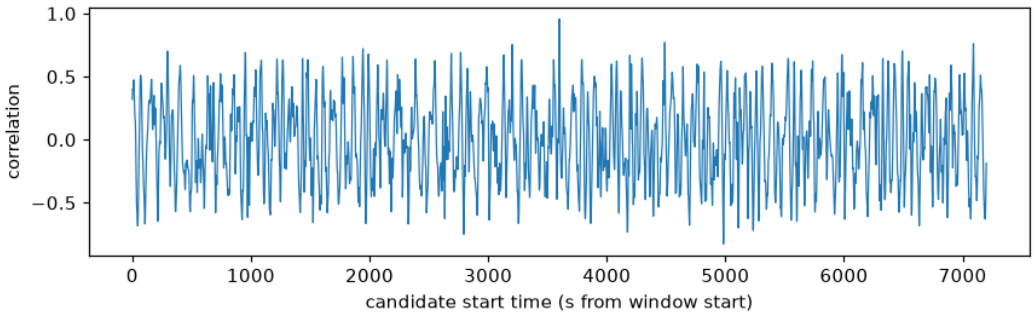
Summary

Estimated recording start (UTC)	2026-01-20T10:00:00.109978+00:00 ± 1.6 s
Grid zone	SIM50
Reasons	• `READOUT_ESTIMATED` — Readout time was estimated from the video (lower confidence). • `MATCH_OK` — The track matches the reference log at one time.
Match	peak correlation 0.957, next peak 0.771, track length 69 s

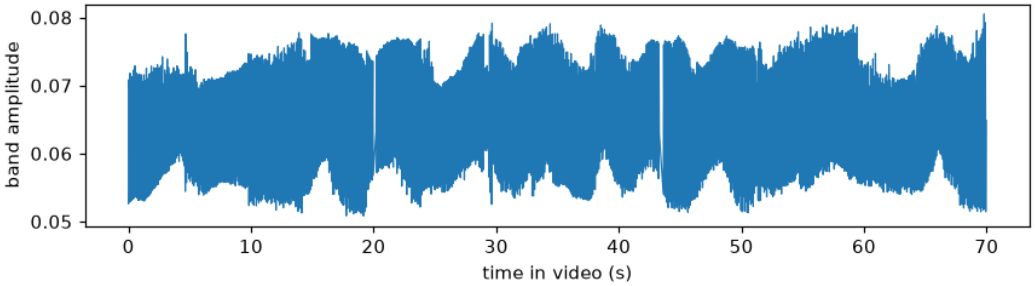
Key plots



Measured grid frequency variation and matched reference.



Correlation against candidate start times.



Flicker band amplitude per frame (fractional modulation).

Limitations

- GRIDPRINT reads the grid frequency from the flicker of mains-powered lights. It is not a deepfake classifier and it is not cryptographic proof.
- A missing signal is never evidence that a video is fake. Many phones hide the flicker ("anti-banding"), and daylight or battery lights do not flicker.
- A time estimate is possible only inside the time range of the reference log for the zone.
- Location output is limited to the grid zone. GRIDPRINT gives no finer location.
- Thresholds come from calibration data. Results outside the calibrated conditions carry more risk of error.
- A CONTRADICTED verdict needs approval by a human reviewer before the report is final.

Provenance

File SHA-256	

	`97bd81ea29c2a33b16465d912f7ac4e526a0425b1cc54b33e27aeb10b46074b9`
Engine version	0.1.0
Config hash	`8f439372b865ee8e`
Thresholds hash	`0161b96f77d3e925`
Job	`aws4-dropped-frames-041757`

Appendix

Continuity events

None.

Quality metrics

axis	rows+
band_snr_db	8.066753447894417
bitrate	490747
bits_per_pixel	0.21453018834932053
camera_motion_px	0.032422965009069944
codec	h264
dft_peak_hz	100.02934823200873
estimators_agree	True
foreground_fraction	0.052121927458033575
frames	2085
height	240
jitter_p50_ms	3.3305269653283176e-07
jitter_p95_ms	3.338056941970535e-07
motion_level_px	0.10989784047152439
nominal_flicker_hz	100.0
opencv_frame_count	2085
phase_mean_flicker_hz	100.02626642175629

readout_s	0.025285469843590924
readout_source	READOUT_ESTIMATED
rotation_metadata_deg	0
segmentation	ok
static_fraction	1.0
timestamp_source	sensor
timing_assumption	sensor_timestamp_is_first_row_exposure_start
track_id	t1
track_samples	66
transfer	bt709
verdict_rule	S13.4
width	320

Stage timings (s)

S1_ingest	0.3257
S2_linear	2.1445
S3_readout	2.0851
S7_rowsignal	2.6154
S4_motion	6.4908
S5_segment	1.006
S6_patches	1.5211
S8_phase	0.7494
S10_match	0.7262